

Quantum Machine Learning Photonic Thermometry: A Finite Mathematical Framework for Deep Geometry Contrast

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Abstract

We present a novel quantum machine learning (QML) paradigm leveraging photonic circuitry arrays to reconstruct high-fidelity, deep geometry contrast in MR thermometry. Traditional MR pulse sequences lack the structural resonance to discern microscopic temperature fluctuations in heterogeneous biological tissue. By introducing Qubit Photonic Coil geometries intertwined with QML-optimized pulse sequences (QML Photonic Therm v1 & v2), we mathematically model and demonstrate superior signal-to-noise (SNR) topologies. Our finite deterministic framework calculates temperature anomalies using quantum topological wave modulations, substantially amplifying local target contrast without triggering noise-floor violations. This methodology paves a clinically viable pathway for ultra-precise neurovascular thermal ablations.

1. Qubit Photonic Circuitry Coil Sensitivity

The complex RF sensitivity $S_c(r, \theta_c)$ of the qubit-coupled photonic array integrates an entanglement amplitude term with standard B1 phase accumulation:

$$S_c(x, y) = e^{-\frac{d}{\lambda_1}} \cdot e^{i(\frac{d}{\lambda_2} + \theta_c)} \left[1.0 + \eta \cdot \Gamma_{noise} \cdot \sin\left(\frac{x}{\lambda_3}\right) \right]$$

where $d = \sqrt{(x-x_c)^2 + (y-y_c)^2}$ is the radial distance from coil center c , and $\eta=0.1$ is the quantum dispersion coefficient.

2. MR Thermometry Probability Density & Base Temperature

The base thermal equilibrium is disrupted by targeted heating, modulated by the qubit topology. Given a radial distance R from the isocenter:

$$T(R) = 37.0 + \Delta T_{max} e^{-R^2/\sigma^2} + 2.0 \sin(5 \cdot R)$$

The sine term represents the distinct ripple resonance mapped by the QML topology.

3. Photonic Contrast and Spin Echo Modulation

Pulse sequence integration relies on altering transverse magnetization (M_{xy}) metrics. For our deep volume sequence (v2):

$$M_{xy}^{(v2)} = \rho_0 (1 - e^{-TR/T_1}) e^{-TE/T_2} \left[\frac{SNR_{qml}}{80.0} \right]$$

For the superficial variant (v1), the denominator shifts to 100.0, limiting the deep contrast but maximizing near-surface resolution.

4. Photonic Temperature Resonance vs Baseline

Thermal Activation Maps with QML Contrast Scaling

